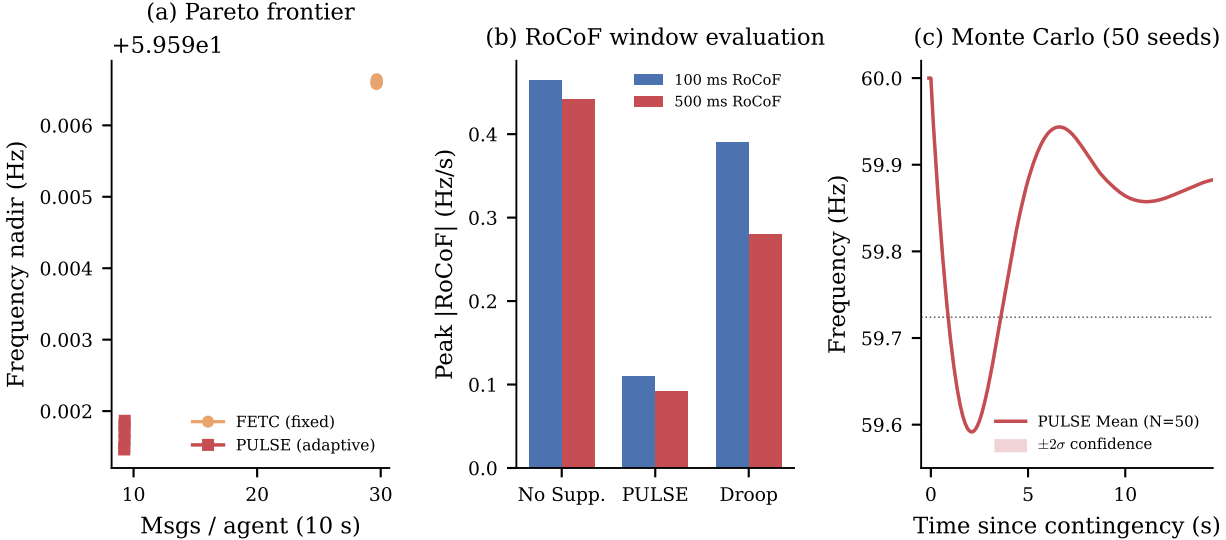

(a) Pareto frontier
+5.959e1
Frequency nadir (Hz)
0.006
0.005
0.004
0.003
0.002
FETC (fixed)
PULSE (adaptive)
10
20
30
Msgs / agent (10 s)
(b) RoCoF window evaluation
100 ms RoCoF
500 ms RoCoF
Peak |RoCoF| (Hz/s)
0.4
0.3
0.2
0.1
0.0
No Supp.
PULSE
Droop
(c) Monte Carlo (50 seeds)
Frequency (Hz)
60.0
59.9
59.8
59.7
59.6
PULSE Mean (N=50)
±2σ confidence
0
5
10
Time since contingency (s)